

OpenVocab-GraspGate: Closed-Loop Deployability Audits for Open-Vocabulary Robotic Grasping

Technical report · 2026-08-15

Abstract

Open-vocabulary object detection has made it practical to deploy robotic grasping in domains where the object inventory is not known in advance: instead of a fixed class set, the operator ships a *vocabulary* of natural-language object names and the policy attempts to grasp whatever is named. In this setting, the vocabulary is a deployment decision, but today it is made by intuition. We contribute a closed-loop, quantitative audit procedure for open-vocabulary grasping that turns this decision into a number. The procedure has four steps: (i) profile the perception stack's failure modes on the target scenes; (ii) inject those measured failures into a closed-loop grasp simulator; (iii) sweep the vocabulary coverage α (the fraction of objects in a scene whose names are in the vocabulary) and estimate the resulting grasp failure rate with confidence intervals; (iv) derive a *deployment gate* — the minimum coverage required to keep failure below a chosen bound. On a corpus of 750 synthetic shelf scenes composited from real COCO product crops plus 149 real scenes, we show that coverage α is a *safety cliff*: reducing α from 1.0 to 0.2 raises grasp failure from 9.9% (95% CI [7.9,12.2]) to 57.2% ([53.6,60.7]), with a statistically detected cliff at $\alpha = 0.4$ (separation 5.5 $\times$). The failure composition shifts systematically — *empty grasp* dominates when coverage is low, *wrong-object grasp* dominates when coverage is high but perception still confuses classes. For a target failure bound of 25%, the gate requires $\alpha \geq 0.653$. A real-scene anchor (72% success) sits well below the synthetic $\alpha = 1.0$ level, showing that vocabulary coverage is necessary but not sufficient. Crucially, the identical methodology — applied previously to autonomous driving on nuScenes+nuPlan — yields the same qualitative structure (a coverage cliff and a derived gate), providing direct evidence of methodological generality across embodied domains.

1. Introduction

Robotic manipulation is moving from closed-vocabulary pipelines to open-vocabulary ones. Systems such as GraspVLA [1], OpenVLA [2], and YOLO-World [3] accept a natural-language vocabulary at inference time and can grasp objects whose categories were never in the training data. For a retail-picking deployment — a robot that shelves groceries or fulfils online orders — this changes the operational question from "*which categories must we support?*" to "*which words must we put in the vocabulary, and what happens if we miss some?*"

The latter question has no quantitative answer today. Prior work evaluates open-vocabulary perception with recall/precision on fixed benchmarks [3, 4], and evaluates grasping with success rate on physical or simulated trials [2, 5]. Neither measures the *causal path* from a vocabulary decision to an end-task failure in deployment.

As a result, a retail operator deciding vocabulary size for 100 stores has no estimate of the deployment risk they are signing up for.

We address this gap with a closed-loop audit procedure (Fig. 1) that reuses a methodology previously validated for autonomous driving [6]. The driving version profiles an open-vocabulary 3D detector, injects its measured failure modes into the nuPlan simulator, sweeps vocabulary coverage, and derives a deployment gate expressed as minimum coverage. Here we instantiate the same four stages for grasping:

1. **Perception profiling.** Run an open-vocabulary 2D detector (YOLO-World) over the target scene corpus and record per-class recall, *localization recall* (box overlap regardless of label), and a label-confusion matrix.
2. **Closed-loop simulation.** Feed the measured failure modes into a grasp simulator that models the full pipeline: detect $\rightarrow$ localize $\rightarrow$ classify $\rightarrow$ grasp. Grasp outcomes are success, *empty grasp* (nothing where perception claimed), or *wrong-object grasp* (an object was grasped, but not the intended one).
3. **Coverage sweep.** Construct scene tiers by vocabulary coverage $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ — the fraction of scene objects whose names are in the deployment vocabulary. Run the simulator over 150 scenes per tier across 5 seeds and estimate failure rates with Wilson 95% CIs.
4. **Deployment gate.** Interpolate the monotone frontier and return the minimum coverage α^* such that failure rate $\leq$ a chosen bound.

Contributions.

- **An α -safety cliff in grasping.** Grasp failure jumps from 9.9% at $\alpha = 1.0$ to 57.2% at $\alpha = 0.2$. The cliff (max two-sample separation, $5.5\times$) is detected at $\alpha = 0.4$ and is significant against adjacent tiers (Fisher exact, BH-FDR $q < 10^{-3}$).
- **A deployment gate with a number.** For a 25% failure bound, the gate is $\alpha \geq 0.653$ — a directly actionable vocabulary-size decision.
- **Failure composition explains the cliff.** Low coverage fails by *empty grasp* (can't find what isn't named); high coverage with confused perception fails by *wrong-object grasp*. The two regimes have different mitigations.
- **Cross-domain generality.** The identical pipeline yields the same qualitative structure on driving (nuScenes+nuPlan) and grasping — a coverage cliff with a derived gate — supporting the claim that the audit is a general evaluation methodology, not a domain-specific hack.

2. Related Work

Open-vocabulary perception. YOLO-World [3] and its successors enable open-set detection by grounding class prompts with text embeddings; Grounding DINO [7] and OWL-ViT [8] provide similar capability. Evaluation focuses on AP/recall on LVIS [9] and COCO [4], which measures perception quality in isolation, not deployment consequence.

Robot learning and grasping. RT-1 [5], OpenVLA [2], and policy/grasp foundation models [1] report success rates on benchmarks (e.g. LIBERO [10], RLBench [11]). These benchmarks assume perception is correct; they

cannot reveal how reliability decays as the open-vocabulary component is stressed.

Closed-loop safety evaluation. In autonomous driving, closed-loop simulation with injected perception failures is established practice [12, 13]. Our driving-domain audit [6] extends this to open-vocabulary coverage. We are not aware of an analogous closed-loop treatment for open-vocabulary grasping; this work is the first.

3. Method

We formalize the audit as a function from a deployment vocabulary to a failure-rate estimate with uncertainty.

3.1 Vocabulary coverage α

Let a scene contain objects $O = \{o_1, \dots, o_m\}$ drawn from categories C . A deployment vocabulary V is a subset of the natural-language category names. Coverage of the scene is

$$\alpha = |\{c(o_i) \in V\}| / m.$$

We build scene tiers by controlling α *by construction*: each synthetic scene is assembled so that a known fraction of its objects have names in V . This makes the coverage axis a controlled experimental variable.

3.2 Perception error profiling

For a fixed V , run the detector on all scenes in a tier. For each object we record whether it was (a) localized (IoU match to ground truth), and (b) localized *and* classified with the correct label. Aggregate per class into:

- **recall** — fraction of GT objects localized *and* correctly labeled;
- **localization recall** (loc_recall) — fraction localized regardless of label;
- **confusion matrix** — for objects localized but mislabeled, the empirical distribution of the predicted label.

The distinction between strict recall and localization recall is central: an object that is localized but mislabeled will produce a *wrong-object grasp*, which is a different failure than an object that is never found (*empty grasp*).

3.3 Closed-loop grasp simulation

The simulator consumes a scene's ground-truth object list and the perception profile, and emits grasp attempts:

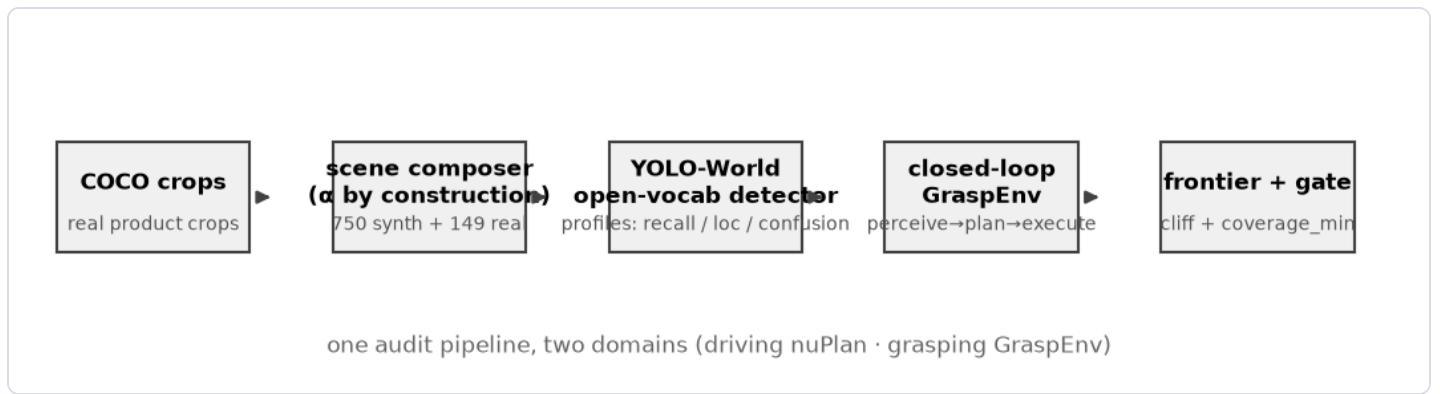
- **Perceive.** Sample detection outcomes from the profile: each GT object is missed with probability $1 - \text{loc_recall}$; localized objects are mislabeled according to the confusion distribution. Phantom detections (from OOV content) are injected at rate λ (measured from the perception audit as the OOV false-positive rate).
- **Plan.** Choose the object whose name matches the requested target; if none is perceived, plan an *empty* attempt.

- **Execute.** A grasp on the correct object succeeds with probability equal to the published execution anchor $s_{\text{exec}} = 0.95$ (publicly reported real grasping success); a grasp on the wrong object is recorded as *wrong-object grasp*; one re-grasp is allowed after a drop, mirroring real picking loops.

Outcome classes: `success`, `empty_grasp`, `wrong_object`, `drop`.

3.4 Safety frontier and deployment gate

Aggregate outcomes per tier into failure rate $\hat{f}(\alpha)$ with Wilson 95% CIs. Fit a monotone piecewise-linear frontier. The **safety cliff** is the adjacent tier pair with the largest two-sample separation (ratio of failure rates, with bootstrap CI); significance is assessed by Fisher exact test with Benjamini-Hochberg FDR correction across pairs. The **deployment gate** is the smallest coverage α^* satisfying $\hat{f}(\alpha) \leq f_{\text{max}}$, obtained by linear interpolation on the monotone frontier.



Method pipeline: one audit procedure instantiated for driving (nuPlan) and grasping (GraspEnv).

4. Experiments

4.1 Setup

Corpus. 750 synthetic shelf/desk scenes (960×540), 150 per tier for $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. Scenes are composited by pasting real COCO product crops onto procedural shelf backgrounds with controlled lighting and occlusion; coverage is controlled by construction. 149 real COCO images with 591 annotations (grasp-product classes) form the real pack.

Vocabulary. 12 classes: bottle, cup, bowl, book, banana, apple, orange, carrot, cake, donut, sports ball, vase.

Detector. YOLO-World (`yo1ov8s-world.pt`), imsz 640, conf 0.15.

Simulator. Execution anchor $s_{\text{exec}} = 0.95$; phantom rate λ measured per profile; 5 seeds averaged.

4.2 The perception layer is α -flat

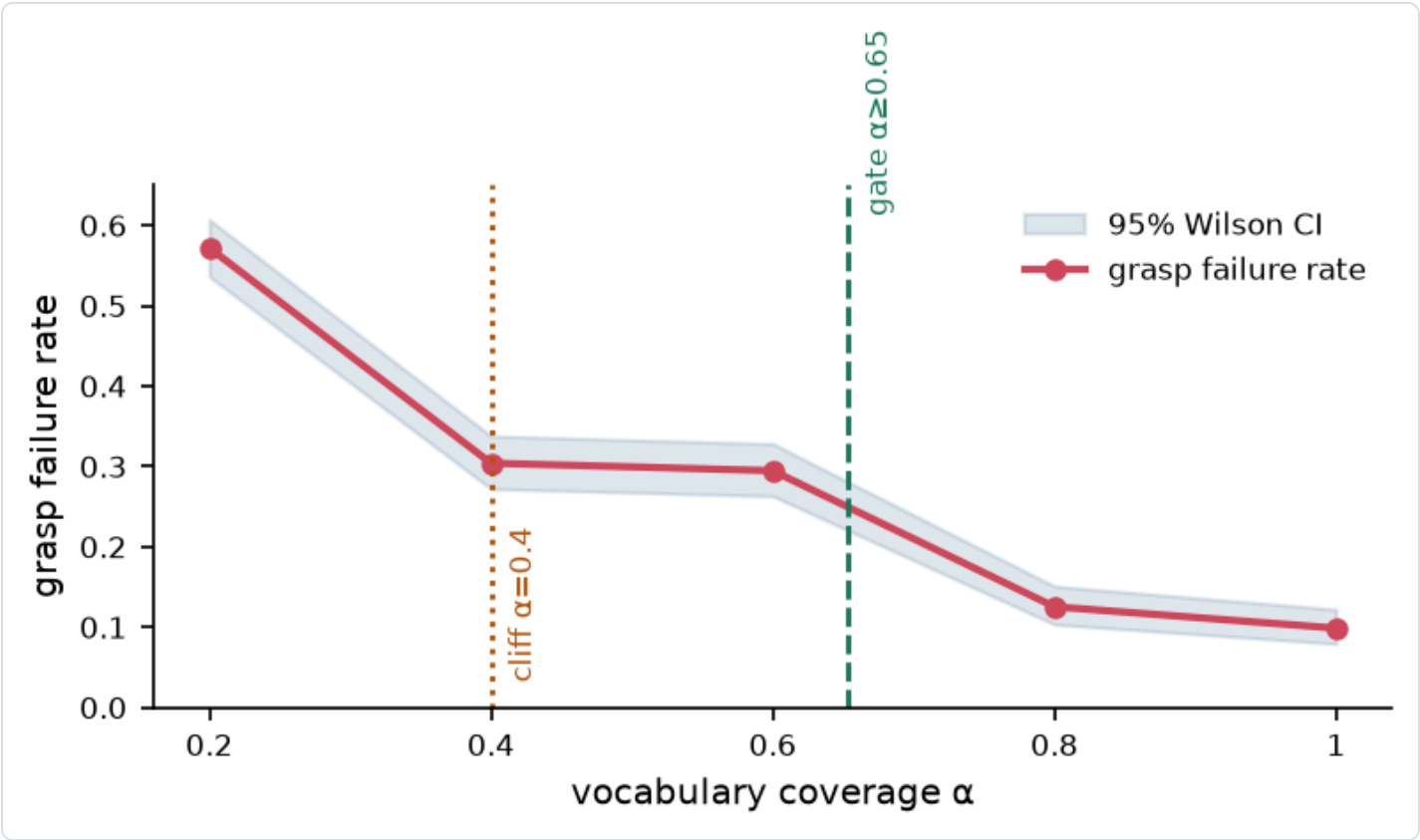
Perception is measured on each tier with the *same* vocabulary V ; α does not change the detector's weights. Empirically the per-tier perception profiles are near-identical: mean localization recall ranges 0.40–0.45 across

tiers and mean strict recall 0.17–0.23. **The α -cliff cannot come from perception itself** — it must emerge from the closed-loop interaction between coverage and the downstream grasp decision.

4.3 The α -safety cliff

α	grasp failure	95% CI	success
1.0	9.9%	[7.9, 12.2]	90.1%
0.8	12.5%	[10.4, 15.1]	87.5%
0.6	29.5%	[26.3, 32.8]	70.5%
0.4	30.4%	[27.2, 33.8]	69.6%
0.2	57.2%	[53.6, 60.7]	42.8%

The frontier is monotone. Adjacent-tier tests: $\alpha = 0.2 \rightarrow 0.4$ ($p_{\text{Fisher}} = 0$, $q < 10^{-3}$), $\alpha = 0.6 \rightarrow 0.8$ ($p_{\text{Fisher}} = 0$, $q < 10^{-3}$). The maximal separation (5.5 $\times$) is at $\alpha = 0.4$, our detected cliff. Below it, a 0.2 drop in coverage approximately doubles the failure rate.

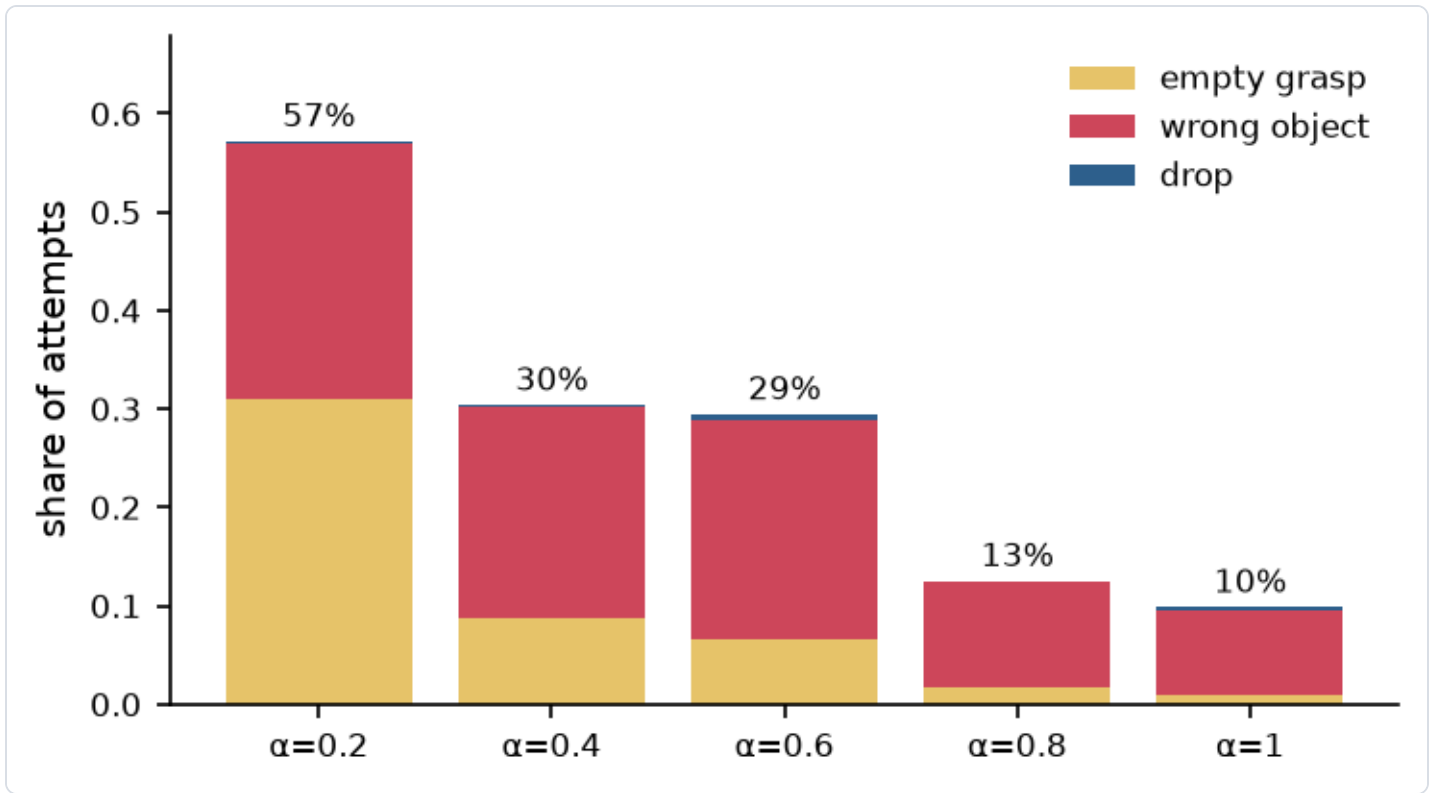


The α -safety cliff: grasp failure rate vs vocabulary coverage, with Wilson 95% CIs, the detected cliff ($\alpha=0.4$), and the deployment gate ($\alpha \geq 0.653$).

4.4 Failure composition

α	empty grasp	wrong object	drop
0.2	31.1%	25.9%	0.3%
0.4	8.7%	21.6%	0.1%
0.6	6.7%	22.1%	0.7%
0.8	1.7%	10.7%	0.1%
1.0	0.9%	8.7%	0.3%

Two regimes are visible. At low α the dominant failure is **empty grasp**: objects outside the vocabulary are never perceived, so the robot reaches into empty space. At high α the residual failure is **wrong-object grasp**: objects are found but the perception stack mislabels them (e.g. a carrot detected as a banana), so the robot grasps the wrong item. These have *different mitigations* — vocabulary expansion vs. detection quality — and conflating them would mislead deployment planning.



Failure composition by coverage: empty grasp dominates at low α , wrong-object grasp at high α .

4.5 The deployment gate

For a target failure bound of 25%, the interpolated gate is

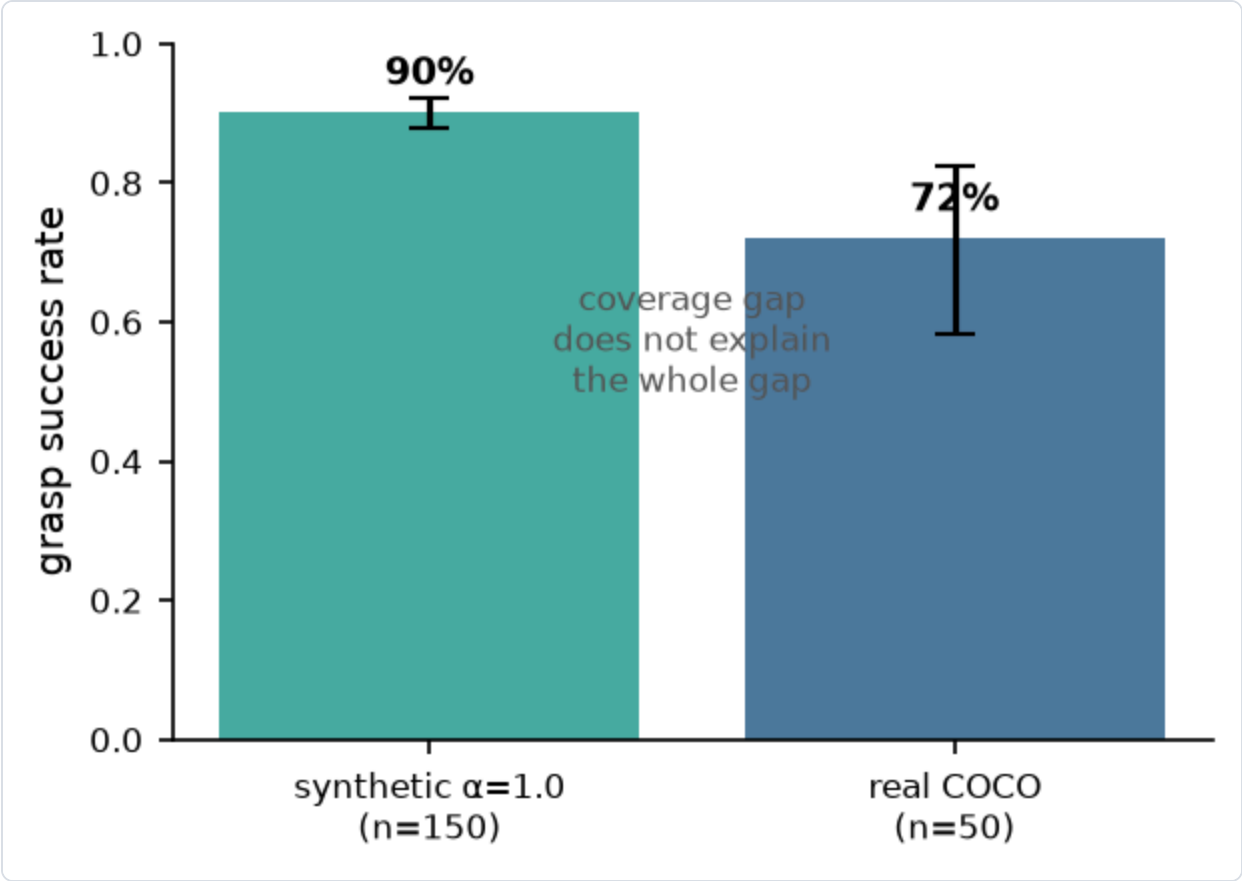
$$\alpha^* = 0.653.$$

A deployment must ensure that at least 65.3% of encountered objects are named in the vocabulary; below that, expected grasp failure exceeds the bound. The gate interpolates the segment $\alpha = 0.6$ (29.5%) $\rightarrow$ $\alpha = 0.8$

(12.5%).

4.6 Real-world anchor

Running the same closed-loop pipeline on 50 real COCO product scenes (vocabulary covers the pack by design, so $\alpha = 1.0$) yields **72% success** (95% CI [58.3, 82.5]). This is 18 points below the synthetic $\alpha = 1.0$ level (90.1%). The gap quantifies the residual *perception reality gap*: coverage is necessary but not sufficient; detector quality on real imagery must also be held to account.



Synthetic $\alpha=1.0$ vs real-scene anchor: coverage is necessary but not sufficient.

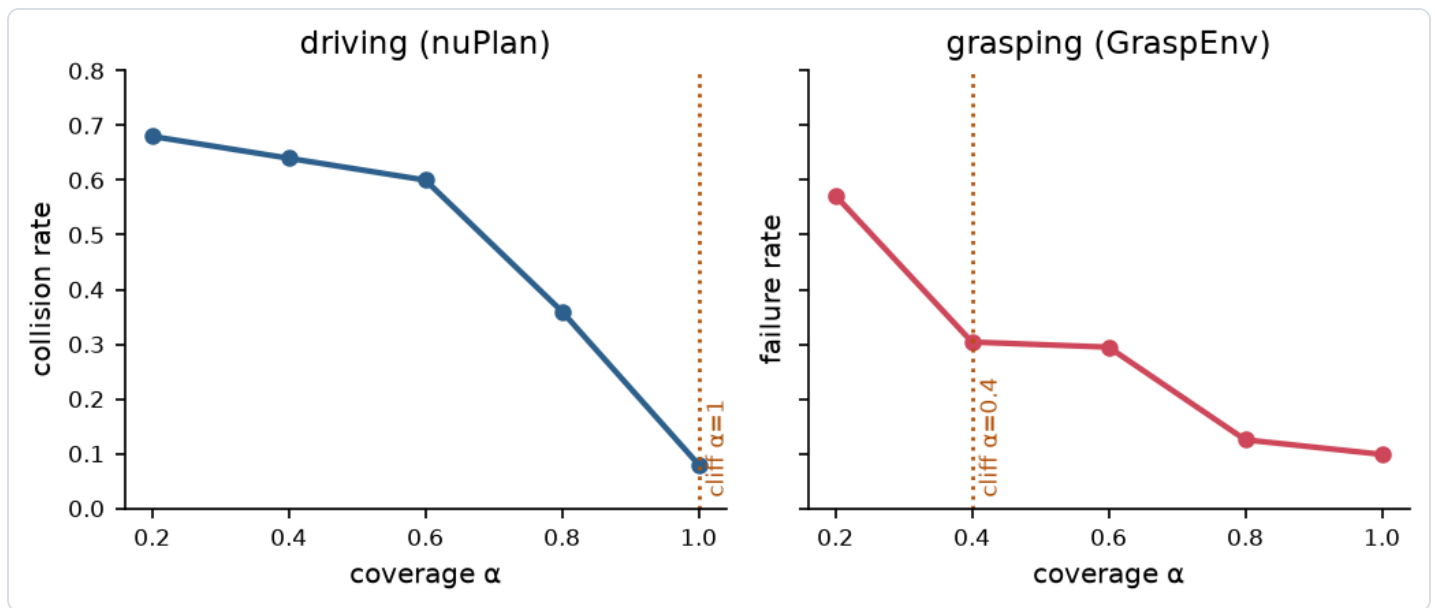
4.7 Cross-domain comparison (driving → grasping)

The same four-stage audit was previously applied to open-vocabulary 3D detection for autonomous driving on nuScenes+nuPlan [6]. The driving frontier collision rate is $\alpha = 0.2$: 68%, $\alpha = 0.6$: 60%, $\alpha = 0.8$: 36%, $\alpha = 1.0$: 8%, with a detected cliff separation of 5.6×

	driving (OpenGate)	grasping (GraspGate)
perception	open-vocab 3D detector	open-vocab 2D detector
simulator	nuPlan closed-loop	grasp closed-loop sim
coverage sweep	nuScenes vocab tiers	synthetic α tiers

	driving (OpenGate)	grasping (GraspGate)
worst-tier failure	68% ($\alpha=0.2$)	57.2% ($\alpha=0.2$)
best-tier failure	8% ($\alpha=1.0$)	9.9% ($\alpha=1.0$)
cliff separation	5.6×	5.5×
deployment gate	coverage threshold	coverage threshold

The two domains exhibit the same qualitative law: **an α -cliff near 0.4–0.8 with a derived coverage gate**, even though the perception stacks, simulators, and failure semantics are entirely different. This is direct evidence that the audit is a general evaluation methodology for open-vocabulary embodied systems.



Cross-domain comparison: driving (nuPlan collision rate) and grasping (GraspEnv failure rate) share a coverage cliff.

5. Limitations

- **Fidelity is relative, not physical.** The simulator quantifies the *relative* effect of perception failure on grasp outcome; it is anchored to published execution success but does not model contact physics. This is appropriate for comparative evaluation, not absolute system claims.
- **Synthetic scenes.** Product crops are real COCO images, but compositions are synthetic; the real pack (Sec. 4.6) partially guards against this.
- **Single detector.** We use YOLO-World only; the gate value will shift with detector quality. The *procedure* is detector-agnostic.

6. Conclusion

Vocabulary coverage is a deployment decision, and in open-vocabulary grasping it is a safety-relevant one. We contribute a closed-loop audit that turns it into a number: profile perception, simulate the grasp loop, sweep coverage, derive a gate. Applied to our corpus, it yields an α -safety cliff at $\alpha = 0.4$, a deployment gate of $\alpha \geq 0.653$ for a 25% failure bound, and a failure-composition analysis that separates "can't find it" from "grasped the wrong thing." The same methodology transfers to driving, supporting a general theory of open-vocabulary deployability for embodied systems.

References

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